Daily Problem Solutions Booklet

Comprehensive Step-by-Step Analytical Solutions

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Integrals

1.1 Easy Integral

Inverse Trigonometric Substitution

Evaluate:

$$\int_0^\infty \tan(\sin^{-1}(e^{-2x}))\, dx$$

Solution: Let $u = e^{-2x}$. Then $du = -2e^{-2x}\, dx \implies dx = -\frac{du}{2u}$. When $x = 0$, $u = 1$. As $x \to \infty$, $u \to 0$. Substitute into the integral:

$$\int_1^0 \tan(\sin^{-1}(u)) \left(-\frac{du}{2u}\right) = \frac{1}{2}\int_0^1 \frac{\tan(\sin^{-1}(u))}{u}\, du$$

Let $\theta = \sin^{-1}(u)$, so $\sin\theta = u$. From a right triangle, the opposite side is u and the hypotenuse is 1, making the adjacent side $\sqrt{1-u^2}$. Thus, $\tan\theta = \frac{u}{\sqrt{1-u^2}}$. The integral becomes:

$$\frac{1}{2}\int_0^1 \frac{\frac{u}{\sqrt{1-u^2}}}{u}\, du = \frac{1}{2}\int_0^1 \frac{1}{\sqrt{1-u^2}}\, du$$

This is a standard integral:

$$\frac{1}{2}\left[\sin^{-1}(u)\right]_0^1 = \frac{1}{2}\left(\sin^{-1}(1) - \sin^{-1}(0)\right) = \frac{1}{2}\left(\frac{\pi}{2} - 0\right) = \frac{\pi}{4}$$

1.2 Medium Integral

Exponential Substitution

If $\alpha = \int_0^1 (e^{9x+3\tan^{-1}(x)}) \left(\frac{12+9x^2}{1+x^2}\right) dx$ where $\tan^{-1}(x)$ takes only principal values, then find the value of:

$$\left(\ln|1+\alpha| - \frac{3\pi}{4}\right)$$

Solution: Let $u = 9x + 3\tan^{-1}(x)$. Taking the derivative with respect to x:

$$\frac{du}{dx} = 9 + \frac{3}{1+x^2} = \frac{9(1+x^2)+3}{1+x^2} = \frac{12+9x^2}{1+x^2}$$

Thus, $du = \left(\frac{12+9x^2}{1+x^2}\right) dx$. Next, update the limits of integration: When $x = 0$, $u = 9(0) + 3\tan^{-1}(0) = 0$. When $x = 1$, $u = 9(1) + 3\tan^{-1}(1) = 9 + 3\left(\frac{\pi}{4}\right) = 9 + \frac{3\pi}{4}$. Substitute u and du into the integral:

$$\alpha = \int_0^{9+\frac{3\pi}{4}} e^u\, du = [e^u]_0^{9+\frac{3\pi}{4}} = e^{9+\frac{3\pi}{4}} - 1$$

We need to evaluate the expression $\ln|1+\alpha| - \frac{3\pi}{4}$:

$$1 + \alpha = e^{9+\frac{3\pi}{4}}$$

$$\ln|1+\alpha| = \ln\left(e^{9+\frac{3\pi}{4}}\right) = 9 + \frac{3\pi}{4}$$

Finally, substitute this back:

$$\left(9 + \frac{3\pi}{4}\right) - \frac{3\pi}{4} = 9$$

1.3 Hard Integral

Evaluating using Parameterized Integrals

Evaluate:

$$\int_0^\infty \frac{(1-x^2)\ln(x)}{1+x^4}\,dx$$

Solution: We can split the integral into two separate integrals:

$$I = \int_0^\infty \frac{\ln(x)}{1+x^4}\,dx - \int_0^\infty \frac{x^2\ln(x)}{1+x^4}\,dx$$

We utilize the general formula for integrals of the form $\int_0^\infty \frac{x^{a-1}}{1+x^b}\,dx = \frac{\pi}{b\sin(a\pi/b)}$. For our denominator $1+x^4$, we have $b = 4$. Differentiating both sides of the general formula with respect to the parameter a gives:

$$\frac{d}{da}\left(\int_0^\infty \frac{x^{a-1}}{1+x^4}\,dx\right) = \int_0^\infty \frac{x^{a-1}\ln(x)}{1+x^4}\,dx = \frac{d}{da}\left(\frac{\pi}{4\sin(a\pi/4)}\right)$$

Using the chain rule on the right-hand side yields:

$$\int_0^\infty \frac{x^{a-1}\ln(x)}{1+x^4}\,dx = -\frac{\pi^2\cos(a\pi/4)}{16\sin^2(a\pi/4)}$$

For the first integral $\int_0^\infty \frac{\ln(x)}{1+x^4}\,dx$, we substitute $a = 1$:

$$\int_0^\infty \frac{\ln(x)}{1+x^4}\,dx = -\frac{\pi^2\cos(\pi/4)}{16\sin^2(\pi/4)} = -\frac{\pi^2(\sqrt{2}/2)}{16(1/2)} = -\frac{\pi^2\sqrt{2}}{16}$$

For the second integral $\int_0^\infty \frac{x^2\ln(x)}{1+x^4}\,dx$, we substitute $a = 3$:

$$\int_0^\infty \frac{x^2\ln(x)}{1+x^4}\,dx = -\frac{\pi^2\cos(3\pi/4)}{16\sin^2(3\pi/4)} = -\frac{\pi^2(-\sqrt{2}/2)}{16(1/2)} = \frac{\pi^2\sqrt{2}}{16}$$

Subtracting the second integral from the first yields the true value:

$$I = -\frac{\pi^2\sqrt{2}}{16} - \left(\frac{\pi^2\sqrt{2}}{16}\right) = -\frac{\pi^2\sqrt{2}}{8}$$

Differentials

2.1 Medium Differential

Riccati Differential Equation

The function $y(x)$ satisfies the differential equation:

$$\frac{dy}{dx} = y^2 - \frac{2}{x}y + \frac{2}{x^2}, \quad x > 0$$

Given the initial condition $y(1) = 2$, find the value of $y(2)$.

Solution: This is a Riccati equation. Let's use the substitution $y(x) = \frac{v(x)}{x}$. Differentiating with respect to x using the quotient rule gives:

$$\frac{dy}{dx} = \frac{v'x - v}{x^2}$$

Substitute y and y' back into the original differential equation:

$$\frac{v'x - v}{x^2} = \left(\frac{v}{x}\right)^2 - \frac{2}{x}\left(\frac{v}{x}\right) + \frac{2}{x^2}$$

Multiply the entire equation by x^2:

$$v'x - v = v^2 - 2v + 2 \implies x\frac{dv}{dx} = v^2 - v + 2$$

This is a separable differential equation:

$$\frac{dv}{v^2 - v + 2} = \frac{dx}{x}$$

Complete the square for the quadratic in the denominator: $v^2 - v + 2 = \left(v - \frac{1}{2}\right)^2 + \frac{7}{4}$. Integrate both sides:

$$\int \frac{dv}{\left(v - \frac{1}{2}\right)^2 + \left(\frac{\sqrt{7}}{2}\right)^2} = \int \frac{dx}{x}$$

$$\frac{2}{\sqrt{7}} \arctan\left(\frac{2v - 1}{\sqrt{7}}\right) = \ln x + C$$

Apply the initial condition $y(1) = 2$. Since $y = \frac{v}{x}$, $v(1) = 1 \cdot 2 = 2$.

$$\frac{2}{\sqrt{7}} \arctan\left(\frac{2(2) - 1}{\sqrt{7}}\right) = \ln(1) + C \implies C = \frac{2}{\sqrt{7}} \arctan\left(\frac{3}{\sqrt{7}}\right)$$

Now, find $y(2)$. Let $v_2 = v(2) = 2y(2)$.

$$\frac{2}{\sqrt{7}} \arctan\left(\frac{2v_2 - 1}{\sqrt{7}}\right) = \ln(2) + \frac{2}{\sqrt{7}} \arctan\left(\frac{3}{\sqrt{7}}\right)$$

Solve for v_2:

$$v_2 = \frac{1}{2} + \frac{\sqrt{7}}{2} \tan\left(\frac{\sqrt{7}}{2}\ln(2) + \arctan\left(\frac{3}{\sqrt{7}}\right)\right)$$

Finally, since $y(2) = \frac{v_2}{2}$:

$$y(2) = \frac{1}{4} + \frac{\sqrt{7}}{4} \tan\left(\frac{\sqrt{7}}{2}\ln(2) + \arctan\left(\frac{3}{\sqrt{7}}\right)\right)$$

2.2 Hard Differential

Linear Second-Order ODE

A spacecraft correcting its orbit has angular error $\theta(t)$. The guidance computer reports:

$$\theta''(t) - 2\theta'(t) + 2\theta(t) = e^t \sin(t)$$

Given the initial conditions $\theta(0) = 1$ and $\theta'(0) = 0$, find the angular error when $t = \frac{\pi}{4}$.

Solution: First, find the homogeneous solution. The characteristic equation is $r^2 - 2r + 2 = 0$. Roots are $r = 1 \pm i$. Thus, $\theta_h(t) = e^t(C_1 \cos t + C_2 \sin t)$. Because the right-hand side $e^t \sin t$ is a solution to the homogeneous equation, we must multiply our guess for the particular solution by t:

$$\theta_p(t) = te^t(A \cos t + B \sin t)$$

To simplify, let $\theta(t) = e^t u(t)$. Then: $\theta' = e^t(u' + u)$ and $\theta'' = e^t(u'' + 2u' + u)$. Substitute into the ODE:

$$e^t(u'' + 2u' + u) - 2e^t(u' + u) + 2e^t u = e^t \sin t \implies u'' + u = \sin t$$

For $u'' + u = \sin t$, the particular solution is $u_p(t) = t(A \cos t + B \sin t)$. $u_p'' = -2A \sin t - At \cos t + 2B \cos t - Bt \sin t$ Substitute u_p and u_p'' into $u'' + u = \sin t$:

$$(-2A \sin t + 2B \cos t) = \sin t \implies A = -\frac{1}{2}, B = 0$$

So, $u_p(t) = -\frac{1}{2}t \cos t$, and $\theta_p(t) = -\frac{1}{2}te^t \cos t$. The general solution is:

$$\theta(t) = e^t \left(C_1 \cos t + C_2 \sin t - \frac{1}{2}t \cos t \right)$$

Apply initial conditions: $\theta(0) = C_1 = 1$. $\theta'(t) = e^t \left(\cos t + C_2 \sin t - \frac{1}{2}t \cos t\right) + e^t \left(-\sin t + C_2 \cos t - \frac{1}{2} \cos t +\right.$ $\theta'(0) = 1(1 + 0 - 0) + 1(0 + C_2 - \frac{1}{2} + 0) = 1 + C_2 - \frac{1}{2} = C_2 + \frac{1}{2}$ Setting $\theta'(0) = 0 \implies C_2 = -\frac{1}{2}$. Thus, the specific solution is:

$$\theta(t) = e^t \left(\cos t - \frac{1}{2} \sin t - \frac{1}{2}t \cos t \right)$$

Evaluate at $t = \frac{\pi}{4}$:

$$\theta\left(\frac{\pi}{4}\right) = e^{\pi/4} \left(\frac{\sqrt{2}}{2} - \frac{1}{2}\frac{\sqrt{2}}{2} - \frac{1}{2}\frac{\pi}{4}\frac{\sqrt{2}}{2} \right) = e^{\pi/4}\frac{\sqrt{2}}{2} \left(1 - \frac{1}{2} - \frac{\pi}{8} \right)$$

$$\theta\left(\frac{\pi}{4}\right) = \frac{\sqrt{2}(4 - \pi)}{16} e^{\pi/4}$$

Derivatives

3.1 Medium Derivative

Trigonometric Simplification

Let $f(x) = \frac{\cos^3(x)\sin(x) - \sin^3(x)\cos(x)}{\sin(4x)\cos(8x)}$. Find $f'\left(\frac{\pi}{3}\right)$.

Solution: First, simplify the numerator of $f(x)$:

$$\cos^3(x)\sin(x) - \sin^3(x)\cos(x) = \sin(x)\cos(x)(\cos^2(x) - \sin^2(x))$$

Using the double angle identities $\sin(2x) = 2\sin(x)\cos(x)$ and $\cos(2x) = \cos^2(x) - \sin^2(x)$:

$$\frac{1}{2}\sin(2x)\cos(2x) = \frac{1}{4}\sin(4x)$$

Substitute this back into $f(x)$:

$$f(x) = \frac{\frac{1}{4}\sin(4x)}{\sin(4x)\cos(8x)} = \frac{1}{4\cos(8x)} = \frac{1}{4}\sec(8x)$$

Now, find the derivative $f'(x)$:

$$f'(x) = \frac{1}{4} \cdot 8\sec(8x)\tan(8x) = 2\sec(8x)\tan(8x)$$

Evaluate the derivative at $x = \frac{\pi}{3}$: The argument becomes $8\left(\frac{\pi}{3}\right) = \frac{8\pi}{3}$. The angle $\frac{8\pi}{3}$ is coterminal with $\frac{8\pi}{3} - 2\pi = \frac{2\pi}{3}$.

$$\sec\left(\frac{2\pi}{3}\right) = -2, \quad \tan\left(\frac{2\pi}{3}\right) = -\sqrt{3}$$

Substitute these values into $f'(x)$:

$$f'\left(\frac{\pi}{3}\right) = 2(-2)(-\sqrt{3}) = 4\sqrt{3}$$

3.2 Hard Derivative

Directional Derivative

Let $f(x,y) = \tan(xy)e^{\sin(x)} + 2y^y - 2$. Find the directional derivative of f along the vector $\vec{v} = 4\hat{\imath} + e\hat{\jmath}$ at the point $(0,3)$.

Solution: First, find the partial derivatives of f with respect to x and y.

$$f_x = \frac{\partial}{\partial x}[\tan(xy)e^{\sin(x)}] = y\sec^2(xy)e^{\sin(x)} + \tan(xy)e^{\sin(x)}\cos(x)$$

Evaluate f_x at $(0,3)$:

$$f_x(0,3) = 3\sec^2(0)e^0 + \tan(0)e^0\cos(0) = 3(1)(1) + 0 = 3$$

$$f_y = \frac{\partial}{\partial y}[\tan(xy)e^{\sin(x)} + 2y^y - 2] = x\sec^2(xy)e^{\sin(x)} + 2\frac{d}{dy}(y^y)$$

To differentiate y^y, let $u = y^y \implies \ln u = y \ln y \implies \frac{u'}{u} = \ln y + 1 \implies u' = y^y(\ln y + 1)$.

$$f_y = x \sec^2(xy) e^{\sin(x)} + 2y^y(\ln y + 1)$$

Evaluate f_y at $(0, 3)$:

$$f_y(0, 3) = 0 + 2(3^3)(\ln 3 + 1) = 54(\ln 3 + 1)$$

The gradient vector at $(0, 3)$ is $\nabla f(0, 3) = \langle 3, 54(\ln 3 + 1)\rangle$. The direction vector is $\vec{v} = \langle 4, e\rangle$. Find the unit vector $\vec{u}$ in the direction of $\vec{v}$:

$$|\vec{v}| = \sqrt{4^2 + e^2} = \sqrt{16 + e^2} \implies \vec{u} = \frac{\langle 4, e\rangle}{\sqrt{16 + e^2}}$$

The directional derivative is the dot product of the gradient and the unit vector:

$$D_{\vec{u}} f(0, 3) = \nabla f(0, 3) \cdot \vec{u} = \frac{3(4) + 54(\ln 3 + 1)(e)}{\sqrt{16 + e^2}} = \frac{12 + 54e(\ln 3 + 1)}{\sqrt{16 + e^2}}$$

Physics & Engineering

4.1 Electric

Capacitor Dielectric Insertion Work

A parallel-plate capacitor of capacitance $C_0 = 10\,\mu\text{F}$ is charged to a potential difference of $V = 20$ V by a battery. The battery is then disconnected. A dielectric slab of dielectric constant $K = 4$ is carefully inserted to completely fill the space between the plates. Calculate the work done by the external agent in inserting the dielectric slab in microjoules (μJ).

Solution: The initial energy U_i stored in the capacitor is:

$$U_i = \frac{1}{2}C_0V^2 = \frac{1}{2}(10 \times 10^{-6}\,\text{F})(20\text{ V})^2 = \frac{1}{2}(10\,\mu\text{F})(400\text{ V}^2) = 2000\,\mu\text{J}$$

Because the battery is disconnected, the charge Q on the plates remains constant.

$$Q = C_0V = (10\,\mu\text{F})(20\text{ V}) = 200\,\mu\text{C}$$

Upon inserting the dielectric, the new capacitance C_f becomes:

$$C_f = KC_0 = 4(10\,\mu\text{F}) = 40\,\mu\text{F}$$

The final energy U_f stored in the capacitor is:

$$U_f = \frac{Q^2}{2C_f} = \frac{(200\,\mu\text{C})^2}{2(40\,\mu\text{F})} = \frac{40000}{80}\,\mu\text{J} = 500\,\mu\text{J}$$

The work done by the external agent W_{ext} is equal to the change in stored energy:

$$W_{\text{ext}} = \Delta U = U_f - U_i = 500\,\mu\text{J} - 2000\,\mu\text{J} = -1500\,\mu\text{J}$$

4.2 Mechanics

Wedge and Block Acceleration

A smooth wedge of mass $M = 4$ kg is placed on a frictionless horizontal floor. A block of mass $m = 1$ kg is kept on the inclined face of the wedge, which makes an angle of 45° with the horizontal. If the system is released from rest, calculate the magnitude of the horizontal acceleration of the wedge in m s^{-2}. Take $g = 9.8$ m s^{-2}.

Solution: Let the horizontal acceleration of the wedge (to the left) be A and the normal reaction between the block and the wedge be N. The equation of motion for the wedge in the horizontal direction is:

$$N\sin(45^\circ) = MA$$

For the block, we work in the non-inertial reference frame of the accelerating wedge. We must apply a pseudo force mA acting horizontally to the right on the block. The forces acting on the block perpendicular to the inclined plane are in equilibrium:

$$N + mA\sin(45^\circ) = mg\cos(45^\circ)$$

Substitute $N = \frac{MA}{\sin(45^\circ)}$ into this equation:

$$\frac{MA}{\sin(45^\circ)} + mA\sin(45^\circ) = mg\cos(45^\circ)$$

Multiply through by $\sin(45^\circ)$:

$$MA + mA\sin^2(45^\circ) = mg\sin(45^\circ)\cos(45^\circ)$$

Since $\sin(45^\circ) = \cos(45^\circ) = \frac{1}{\sqrt{2}}$, we have $\sin^2(45^\circ) = \frac{1}{2}$ and $\sin(45^\circ)\cos(45^\circ) = \frac{1}{2}$. Substitute the known values ($M = 4$ kg, $m = 1$ kg, $g = 9.8\ \mathrm{m/s^2}$):

$$4A + 1(A)\left(\frac{1}{2}\right) = 1(9.8)\left(\frac{1}{2}\right)$$

$$4.5A = 4.9 \implies A = \frac{4.9}{4.5} = \frac{49}{45} \approx 1.088\ \mathrm{m\ s^{-2}}$$

4.3 Quantum

Heisenberg Uncertainty Principle

An electron with mass $m = 9.1 \times 10^{-31}$ kg is moving with a speed of $v = 300\ \mathrm{m\ s^{-1}}$. If its speed is measured with an accuracy of 0.01%, calculate the minimum theoretical uncertainty in its position in millimeters (mm). Take $h = 6.63 \times 10^{-34}$ J s and $\pi \approx 3.1416$.

Solution: First, determine the uncertainty in the velocity (Δv):

$$\Delta v = 0.01\% \text{ of } 300\ \mathrm{m/s} = \left(\frac{0.01}{100}\right) \times 300 = 0.03\ \mathrm{m\ s^{-1}}$$

Next, calculate the uncertainty in momentum (Δp):

$$\Delta p = m\Delta v = (9.1 \times 10^{-31}\ \mathrm{kg})(0.03\ \mathrm{m\ s^{-1}}) = 2.73 \times 10^{-32}\ \mathrm{kg\ m\ s^{-1}}$$

The Heisenberg uncertainty principle states that $\Delta x \Delta p \geq \frac{h}{4\pi}$. The minimum theoretical uncertainty in position (Δx_{min}) is:

$$\Delta x_{\text{min}} = \frac{h}{4\pi\Delta p}$$

Substitute the given values:

$$\Delta x_{\text{min}} = \frac{6.63 \times 10^{-34}}{4(3.1416)(2.73 \times 10^{-32})}$$

Calculate the denominator:

$$4 \times 3.1416 \times 2.73 \times 10^{-32} = 12.5664 \times 2.73 \times 10^{-32} = 34.306272 \times 10^{-32}$$

Now divide:

$$\Delta x_{\text{min}} = \frac{6.63 \times 10^{-34}}{34.306272 \times 10^{-32}} \approx 0.193259 \times 10^{-2}\ \mathrm{m}$$

Convert the result from meters to millimeters:

$$0.193259 \times 10^{-2}\ \mathrm{m} = 1.93259 \times 10^{-3}\ \mathrm{m} \approx 1.93\ \mathrm{mm}$$